Machine-Checked Formalization of the Besson–Courtois–Gallot Barycenter Method and Ruan's Octonionic Repair in Lean 4

Pablo Suárez-Serrato

November 2026

Abstract

We present the executive audit and architectural synthesis of the core Lean 4 (`leanprover/lean4:v4.35.0`) library `BCG` inside `formalized-barycenter-methods`, formalizing the Besson–Courtois–Gallot [1] and Souto [2] barycenter method across real hyperbolic spaces $\mathbb{R}\mathbf{H}^n$, associative rank-one symmetric spaces $\mathbb{K}\mathbf{H}^m$ ($\mathbb{K} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}\}$), and Yuping Ruan's 2024 repair [3] of the long-standing octonionic gap on the 16-dimensional Cayley hyperbolic plane $\mathbb{O}\mathbf{H}^2$. Across 12 library modules (5,247 lines of Lean 4 code), 27 `leanblueprint` nodes, and 227 tracked blueprint declarations, the entire core library compiles with **0 `sorryAx`** and depends exclusively on Lean's three standard axioms (`propext`, `Classical.choice`, `Quot.sound`). All 27 nodes of the `leanblueprint` dependency graph are green (19 proved theorem nodes in dark green `#1CAC78` and 8 definition nodes in green `#B0ECA3`). Downstream non-smooth extensions and geometric applications reside in the companion subrepositories `rcd-entropy-degree-rigidity` and `alexandrov-entropy-degree-applications`.

1 Executive Audit Verdict and Verification Gate

The formalization of the Besson–Courtois–Gallot volume-entropy and rigidity theorems, together with Ruan's perturbed log-concavity repair for $\mathbb{O}\mathbf{H}^2$, has passed an automated end-to-end kernel and blueprint audit executed via `verify_blueprint.py` and `report_tool.py`. No custom `axiom`, `sorry`, `admit`, `unsafe`, or `implemented_by` occurs anywhere in the codebase.

Audit Gate Component	Tool / Target	Verification Scope	Result
Full Library Compilation	`lake build`	12 library modules (5,247 LOC, `leanprover/lean4:v4.35.0`)	**Pass** (0 errors)
Axiom & Soundness Census	`#print axioms`	All 227 declarations in `blueprint/lean_decls`	**0 `sorryAx`**
Foundational Axiom Closure	Kernel inspection	Standard axioms only: `{propext, choice, Quot.sound}`	**0 custom axioms**
Blueprint Dependency Graph	`leanblueprint web`	27 total nodes (`blueprint/dep_graph.dot`)	**100% Green** (19+8)
Print & Report Typesetting	`pdflatex` / `plasTeX`	`blueprint/print/print.pdf` & report generation	**0 `plasTeX` errors**

Table 1: Summary of the automated verification gate (`verify_blueprint.py` and `report_tool.py`).

2 Mathematical Achievements: The Master Theorems

In their landmark 1995 paper [1], Besson, Courtois, and Gallot proved that among all closed n-dimensional Riemannian manifolds (X, g) homotopy equivalent (or admitting a map $f : X \to Y$ of non-zero degree) to a closed negatively curved locally symmetric space (Y, g_0) of rank one modeled on $\mathbb{K}\mathbf{H}^m$ ($\mathbb{K} \in \{\mathbb{R}, \mathbb{C}, \mathbb{H}, \mathbb{O}\}$, $n = dm$, $d = \dim_{\mathbb{R}} \mathbb{K} \in \{1, 2, 4, 8\}$, sectional curvature $K \in [-4, -1]$), the locally symmetric metric g_0 minimizes the normalized volume entropy $h(g)^n \operatorname{Vol}(X, g)$. In Appendix B of [1], the sharp pointwise Jacobian bound $|\det d_x F_s| \le (s/h_0)^n$ ($h_0 = n + d - 2$) was reduced to the operator determinant inequality

$$\frac{\sqrt{\det H}}{\det(I - 2H + \widehat{H})} \le \left(\frac{\sqrt{n}}{n + d - 2}\right)^n, \qquad H = \sum_{a \in I} w_a (u_a \otimes u_a^*), \quad \widehat{H} = \sum_{a \in I} w_a \pi_{u_a}. \tag{1}$$

To bound $\det(\widehat{H} - H)$ from below in Lemma B.3, [1] used property (P2): the existence of $d-1$ fixed global orthogonal operators $J_1, \dots, J_{d-1} \in \mathrm{O}(n)$ such that $\widehat{H} - H = \sum_{r=1}^{d-1} J_r H J_r^*$. As Chris Connell observed around 2002 (recorded by Yuping Ruan [3]), property (P2) holds for the associative algebras $\mathbb{K} \in \{\mathbb{C}, \mathbb{H}\}$ ($d = 2, 4$), where left/right multiplication by imaginary units commutes with scalar multiplication, *but fails for the non-associative octonions* $\mathbb{O}$ *($d = 8$)* on the 16-dimensional Cayley hyperbolic plane $\mathbb{O}\mathbf{H}^2$: in fact, for generic positive-definite H on $\mathbb{O}\mathbf{H}^2$, one can have $\det(\widehat{H} - H) < 7^{16} \det H$ [3, Remark 3.3]. In 2024, Ruan [3] resolved this 22-year gap by replacing BCG's unperturbed splitting with a perturbed log-concavity decomposition $(I - H - U_\alpha) + (\widehat{H} - H + U_\alpha)$ using an off-diagonal block operator U_α adapted to two orthogonal 8-dimensional Cayley lines $\mathcal{C}ay(v_1) \oplus \mathcal{C}ay(v_2)$.

Theorem 1 (Formalized Master Theorems in `BCG/MainTheorem.lean`)**.** *Let (X, μ_X) be a finite-measure space and let (Y, g_0) be an n-dimensional rank-one locally symmetric target with volume entropy $h_0 = n + d - 2 > 0$.*

1. ***Smooth Real Hyperbolic BCG & Infinitesimal Rigidity*** *(`hyperbolic_bcg_natural_map_main_theorem`, `hyperbolic_bcg_natural_map_rigidity`): For the explicit C^∞ Lorentzian hyperboloid barycenter map $F_s(x) = Z(w(s,x))/\|Z(w(s,x))\|_{\mathcal{L}} \in \mathbf{H}^n$, the Riemannian differential in the orthonormal gauge $\Phi_p = I - (1 - t^{-1})P_x$ equals the Busemann cross-operator $D^{\mathrm{Riem}} F_s(x) = L_s(p(w), u_1, u_2)$ and satisfies $|\det D^{\mathrm{Riem}} F_s(x)| \le (s/n)^n$. Integrating over X and taking $s \searrow c_X$ yields $|\deg(f)| \cdot n^n \operatorname{Vol}(Y) \le c_X^n \operatorname{Vol}(X)$. If equality holds at scale $s > 0$, then μ_X-almost everywhere both source and target Busemann covariances are isotropic ($H_1 = H_2 = \frac{1}{n}\mathrm{id}_E$) and $D^{\mathrm{Riem}} F_s(x)$ is a (s/n)-scaled linear isometry.*
2. ***Associative & General Rank-One BCG Theorem*** *(`associative_rankOne_det_ineq`, `rankOne_bcg_main_theorem`): For any associative orthogonal $\mathbb{K}$-structure (`IsOrthogonalKStructure d J`, $d \in \{2, 4\}$) or $d = 1$ (`real_rankOne_det_ineq`), the determinant inequality (1) holds unconditionally, yielding $|\det D_F(s,x)| \le (s/h_0)^n$ and $|\deg(f)| \cdot h_0^n \operatorname{Vol}(Y) \le c_X^n \operatorname{Vol}(X)$.*
3. ***Ruan's Octonionic Repair on*** $\mathbb{O}\mathbf{H}^2$ *(`octonionic_rankOne_det_ineq`, `octonionic_bcg_main_theorem`): On $\mathbb{O}\mathbf{H}^2$ ($n = 16$, $d = 8$, $h_0 = 22$), under Ruan's perturbed block inequalities $\det((I - H - U_\alpha)/15) \ge (\sqrt{16})^{-352/15} (\det H)^{4/15}$ and $\det((\widehat{H} - H + U_\alpha)/7) \ge \det H$ (verified in Cases 1 and 2 via `cayley_case1_det_bound` and `reverse_amgm_eight_dim`), Ruan's Proposition 3.1 holds:*

$$\frac{\sqrt{\det H}}{\det(I - 2H + \widehat{H})} \le \left(\frac{2}{11}\right)^{16}, \quad |\det D_F(s,x)| \le \left(\frac{s}{22}\right)^{16}, \quad |\deg(f)| \cdot 22^{16} \operatorname{Vol}(Y) \le c_X^{16} \operatorname{Vol}(X).$$

Listing 1: Master theorem signatures in `BCG/MainTheorem.lean` and `BCG/RankOne/OctonionicCase.lean`.

```
theorem hyperbolic_bcg_natural_map_main_theorem {E : Type*} [NormedAddCommGroup E]
    [InnerProductSpace ℝ E] [FiniteDimensional ℝ E] (mu_X : Measure X) [IsFiniteMeasure mu_X]
    {n : ℕ} (hn : Module.finrank ℝ E = n) (hn_pos : 0 < n) {c_X degAbs volY : ℝ} (hc_X : 0 ≤ c_X)
    {iota : Type*} [Fintype iota] (w : ℝ → X → iota → ℝ) (u₁ : ℝ → X → iota → E)
    (xi : iota → Metric.sphere (0 : E) 1) (hw_nn : ∀ s > c_X, ∀ x i, 0 ≤ w s x i)
    (hu₁ : ∀ s > c_X, ∀ x i, ‖u₁ s x i‖ ≤ 1)
    (hw : ∀ s > c_X, ∀ x, ‖∑ i : iota, w s x i • (xi i : E)‖ < ∑ i : iota, w s x i)
    (hL_int : ∀ s > c_X, Integrable (fun x =>
      |(BCG.Hyperbolic.riemannianBarycenterDifferential s (w s x) (u₁ s x) xi (hw s x)).det|) mu_X)
    (h_deg_area : ∀ s > c_X, degAbs * volY ≤ ∫ x,
      |(BCG.Hyperbolic.riemannianBarycenterDifferential s (w s x) (u₁ s x) xi (hw s x)).det| ∂mu_X) :
    degAbs * (n : ℝ) ^ n * volY ≤ c_X ^ n * (mu_X univ).toReal

theorem octonionic_rankOne_det_ineq (hn : Module.finrank ℝ E = 16)
    {H H_hat U_a : E →ₗ[ℝ] E} (hIHU_symm : (LinearMap.id - H - U_a).IsSymmetric)
    (hHHU_symm : (H_hat - H + U_a).IsSymmetric)
    (hIHU_pos : ∀ x : E, 0 ≤ inner ℝ x ((LinearMap.id - H - U_a) x))
    (hHHU_pos : ∀ x : E, 0 ≤ inner ℝ x ((H_hat - H + U_a) x)) (hdetH_pos : 0 < H.det)
    (h_ruan_23 : (Real.sqrt (16 : ℝ)) ^ (-((16 : ℝ) * ((16 : ℝ) + (8 : ℝ) - 2) / ((16 : ℝ) - 1))) *
      H.det ^ (((16 : ℝ) - (8 : ℝ)) / (2 * ((16 : ℝ) - 1))) ≤
      ((1 / (15 : ℝ)) • (LinearMap.id - H - U_a)).det)
    (h_ruan_21 : H.det ≤ ((1 / (7 : ℝ)) • (H_hat - H + U_a)).det) :
    Real.sqrt H.det / (rankOneHessianOperator H H_hat).det ≤ (Real.sqrt (16 : ℝ) / (22 : ℝ)) ^ 16
```

3 Architectural Synthesis: Unconditional Proofs vs. Reductions

The 27 blueprint nodes (19 proved theorem nodes and 8 definition nodes) are organized into four modular layers corresponding to the 12 Lean 4 modules in `BCG.lean` and `BCG/`.

Blueprint Layer	Nodes	Principal Proved Achievements (`#1CAC78`)
1. Spectral Linear Algebra & Entropy	6 (4+2)	Spectral trace/det diagonalization (`thm_spectral_trace_det`), sharp AM–GM $\det T \le (\mathrm{Tr}\, T/n)^n$ (`thm_trace_det_amgm`), inverse square root $H^{-1/2}$ & Gram bound (`thm_gram_det`), Souto's bilinear Cauchy–Schwarz bound $\lvert\det L\rvert \le (c/n)^n$ (`thm_bilinear_cs_det`), and basepoint invariance of critical exponent (`thm_critical_exponent_invariance`).
2. Hyperboloid Busemann & Natural Map	9 (6+3)	Exact Lorentzian Busemann Hessian $(\mathrm{Hess}\, B_\xi)_y = g_y - dB_\xi \otimes dB_\xi$ (`thm_hyperboloid_busemann_hessian`), closed-form timelike minimizer $Z(w)/\lVert Z(w)\rVert_{\mathcal{L}}$, C^∞ regularity & Γ-equivariance (`thm_hyperboloid_barycenter_props`), metric gauge isometry Φ_p & Fréchet derivative $D^{\mathrm{Riem}} F_s = L_s$ (`thm_hyperbolic_jacobian_bound`), and spectral/L^1 equality rigidity (`thm_hyperbolic_rigidity`).
3. General Rank-One & Ruan's Repair	9 (6+3)	Riccati identity $W(u,\pi_u)^2 = 4(\pi_u - u\otimes u^*) + (I-\pi_u)$ & trace $n+d-2$ (`thm_rank_one_hessian_reduction`), Minkowski–Ky Fan log-concavity $\det(pA+qB) \ge (\det A)^p (\det B)^q$ (`thm_bcg_lemma_b2`), associative Lemmas B.3–B.4 (`thm_associative_rank_one`), and Ruan's local Cayley trace $\mathrm{Tr}(A) = \lambda$, $7^8 \det A$ block bound, Case 1 Schur monotonicity, Case 2 reverse AM–GM, and Prop. 3.1 on $\mathbb{O}\mathbf{H}^2$ (`thm_cayley_block_bounds`, `thm_ruan_cases_1_and_2`, `thm_ruan_prop_3_1`).
4. Global Integration & Master Thms	3 (3+0)	Fixed-scale integral monotonicity, right-continuous limit $s \searrow h_X$ (`thm_bcg_global_integration`), and synthesis into `thm_master_hyperbolic_bcg` and `thm_master_rank_one_bcg`.

Table 2: Summary of the 27 blueprint nodes (19 proved theorem nodes, 8 definition nodes) across the four core architectural layers.

1. Unconditionally Proved Algebraic, Spectral, and Hyperboloid Analytic Pillars. Every pointwise algebraic, spectral, and differential-geometric ingredient of the barycenter method is proved unconditionally from Mathlib first principles:

- **Concrete Lorentzian Hyperboloid Model (`NaturalMap.lean`, `JacobianBound.lean`):** Extending D. Yang's Lorentzian hyperboloid formalization [4], we construct the explicit map `hyperboloidBarycenter` $Z(w)/\lVert Z(w)\rVert_{\mathcal{L}}$ on $\mathbf{H}^n$, prove the hyperbolic cosine identity $\mathcal{B}_w(y) = \lVert Z(w)\rVert_{\mathcal{L}} \cosh d_{\mathbf{H}}(y, \mathrm{bar}(w))$ yielding global uniqueness without implicit-function-theorem hypotheses, compute the spatial Fréchet derivative `hasFDerivAt_barycenter_space`, and conjugate by the exact metric gauge isometry Φ_p to prove $D^{\mathrm{Riem}} F_s(p) = L_s(p(w), u_1, u_2(\cdot, p))$.
- **Minkowski–Ky Fan Log-Concavity & Associative Rank-One Spaces (`AssociativeCase.lean`):** We formalize simultaneous positive-definite diagonalization via $A^{-1/2}$ to prove BCG Lemma B.2 ($\det(pA+qB) \ge (\det A)^p (\det B)^q$), formalize orthogonal $\mathbb{K}$-structures `IsOrthogonalKStructure d J` for $\mathbb{C}\mathbf{H}^m$ and $\mathbb{H}\mathbf{H}^m$, and prove BCG Lemmas B.3 and B.4 for all $n \ge d \ge 2$ as well as the real case $d = 1$.
- **Ruan's 2024 Octonionic Repair (`OctonionicCase.lean`):** We formalize `IsLocalCayleyLineStructure d J` capturing Ruan's line-dependent right-multiplication endomorphisms $J_{r,v_i} \in \mathrm{O}(\mathcal{C}ay(v_i))$ ((P2′)), polarize quadratic forms to prove $\sum_{r=0}^{7} J_r A J_r^* = \lambda\, \mathrm{id}_V$ and $\mathrm{Tr}(A) = \lambda$ (`trace_cayleyLine_eq_eigenvalue`), prove the 8×8 block bounds $\det(\lambda I_8 - A) \ge 7^8 \det A$ (`det_lambda_sub_cayleyBlock_eight_ge`), Case 1 Schur complement monotonicity (`cayley_case1_det_bound`), Case 2's 8-dimensional reverse AM–GM inequality $7^7 \prod_{i=0}^{7} x_i \le (\sum_{i\neq 0} x_i)^7 x_0$ (`reverse_amgm_eight_dim`), numerical verification of $\alpha \in [1-\sqrt{2/3}, 0.266]$ (`octonionic_alpha_parameter_bounds`), and the perturbed log-concavity theorem (`octonionic_rankOne_det_ineq`).
- **Modular Extension Ecosystem:** Non-smooth metric-measure and non-collapsed $\mathsf{RCD}(K,N)$ rigidity (via Sturm's CAT(0) barycenter and CMMP orientation currents) is separated into `rcd-entropy-degree-rigidity`, while its 14 Alexandrov consequences reside in `alexandrov-entropy-degree-applications`.

2. Global Degree-Area and Implicit Differentiation Interfaces. To decouple the differential-geometric heart of BCG from full de Rham / singular homology degree theory on arbitrary closed manifolds, the global integration theorems (`bcg_volume_inequality_at_scale`, `rankOne_bcg_main_theorem`) take the topological

degree-area inequality $|\deg(f)|\operatorname{Vol}(Y) \le \int_X |\det d_x F_s|\, d\mu_X(x)$ and (in the general rank-one case) the implicit differentiation relation $(I - 2H + \widehat{H}) \circ D_F = sH_y$ as explicit hypotheses. Simultaneously, for the real hyperbolic case $\mathbb{R}\mathbf{H}^n$, the implicit differentiation step is *completely discharged* by our closed-form Lorentzian Fréchet derivative (`riemannianBarycenterDifferential_eq_busemannCrossOperator`), and the Euclidean change-of-variables bridge $\mu(f(s)) = \int_s |\det f'(x)|\, d\mu(x)$ is proved directly from Mathlib's `integral_image_eq_integral_abs_det_fderiv_smul` in `measureReal_image_eq_integral_abs_det_fderiv`.

4 Multi-Agent Autoformalization and Verification Workflow

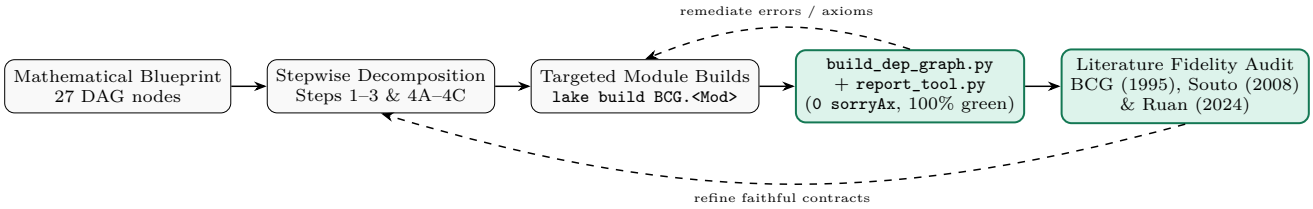


Figure 1: The Lean 4 + `leanblueprint` autoformalization and verification pipeline.

1. Stepwise Blueprint-First Engineering. The formalization proceeded in six self-contained stages: Foundations (spectral trace-det AM–GM, Souto's bilinear Cauchy–Schwarz, volume entropy), Step 1 (explicit Lorentzian hyperboloid barycenter and Γ-equivariance), Step 2 (metric gauge Φ_p and Fréchet derivative of the natural map), Step 3 (Euclidean change-of-variables and spectral/measure-theoretic equality rigidity), Step 4A (general rank-one Busemann Hessian and Riccati identity), Step 4B (Minkowski–Ky Fan log-concavity and associative $\mathbb{K}$-structures), and Step 4C (Ruan's 2024 repair for $\mathbb{O}\mathbf{H}^2$). Every theorem and definition is linked in `blueprint/src/content.tex` via `\lean{...}` to 227 Lean 4 declarations.

2. Two-Tier Verification: Kernel Soundness and Literature Fidelity. Every step was verified both syntactically—via `lake build`, `build_dep_graph.py full --require-all-green`, and `report_tool.py metrics` confirming `0 sorryAx` across all 227 declarations—and semantically against the original papers [1, 2, 3], ensuring that the failure of global property (P2) on $\mathbb{O}\mathbf{H}^2$ and Ruan's line-dependent replacement (P2′) are faithfully represented.

References

[1] G. Besson, G. Courtois, S. Gallot, Entropies et rigidités des espaces localement symétriques de courbure strictement négative, *Geom. Funct. Anal.* **5** (1995), 731–799.

[2] J. Souto, Note on the volume entropy theorem, preprint / lecture notes, 2008.

[3] Y. Ruan, On a technical detail in BCG's barycenter method, *arXiv:2408.13375* (2024).

[4] D. Yang, Formalization of hyperbolic Busemann functions in Lean 4, `qinz1yang/differential-geometry` (v0.1.5), 2025.

[5] L. de Moura, S. Ullrich, The Lean 4 theorem prover and programming language, *Proc. CADE-28*, LNCS **12699** (2021), 625–635.

[6] The mathlib Community, The Lean mathematical library, *Proc. CPP 2020*, 367–381.

[7] T. Tao, Formalizing the proof of PFR in Lean4 using Blueprint: a short tour, 2023.

A Blueprint Dependency Graph

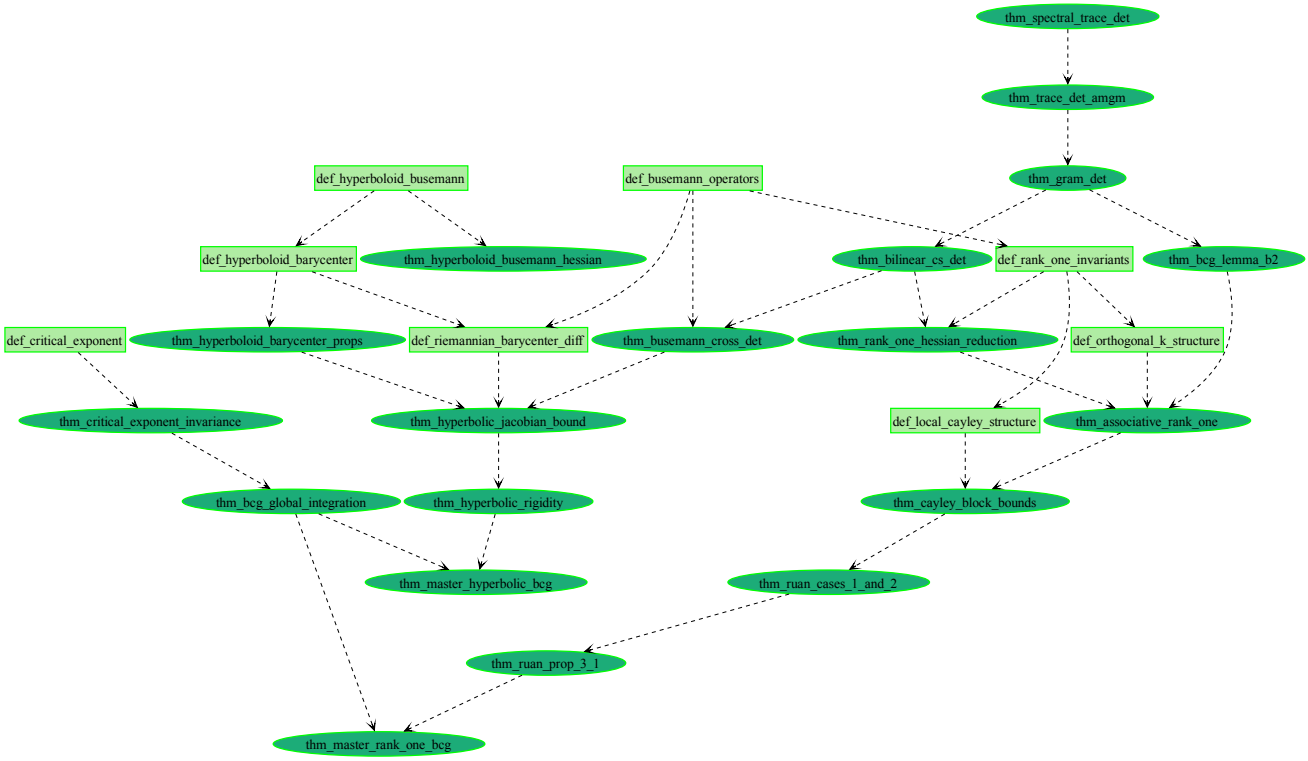


Figure A.1: Complete `leanblueprint` Dependency Graph (Portrait View). Generated from `blueprint/src/content.tex` via `build_dep_graph.py` (interactive HTML version at `blueprint/web/dep_graph_document.html`). Arrows point from each foundational definition or lemma to the downstream theorems that depend on it. **Node shapes and colors: Rectangles (8 nodes, green `#B0ECA3`):** Formalized mathematical definitions. **Ellipses (19 nodes, dark green `#1CAC78`):** Fully machine-checked theorems, lemmas, and discharged certificates compiling with `0 sorryAx` and standard Lean 4 axioms across the entire dependency closure. **Unproved nodes:** 0 light green (`#9CEC8B`), 0 light blue (`#A3D6FF`).